MECH321 Dynamics of Engineering Systems

Lecture 02A: Partial Fractions

Inverse Laplace transform support for Lecture 2

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Partial Fractions for Inversion

Inverse transform of rational $F(s)$

Use partial fractions when rational $F(s)$ does not match a single transform pair.

Rewrite $F(s)$ as an equivalent sum of simpler fractions.

Invert each term using a table pair, then add to obtain $f(t)$.

Connection to Lecture 2

Required self-study before Tutorial 2: slides 2–4, 6 and 7. Slide 5 is an extension.

Partial-fraction method

1 Check before expanding

Cancel common factors. If the numerator degree is not lower, **divide first**. Keep the polynomial part separately.

2 Write every required term

Factor the denominator over the reals. Its factors determine the denominators and the number of unknown coefficients.

3 Find the coefficients

For a simple pole, multiply by its factor, cancel, then evaluate at its root. For repeated poles or a quadratic, match coefficients instead.

4 Invert and check

Use linearity to invert each term. Keep its numerical coefficient. Recombine the fractions to check the original numerator.

Expansion patterns

Distinct linear factors: one constant for each factor

$$\frac{A}{s+a}+\frac{B}{s+b}, \quad a \neq b$$

Repeated linear factor: **include every power up to its multiplicity**

$$\frac{A}{s+a}+\frac{B}{(s+a)^2}+\frac{C}{(s+a)^3}$$

Irreducible quadratic: **allow a linear numerator**

$$\frac{As+B}{s^2+as+b}$$

Example table pair: the coefficient stays outside

$$\mathcal{L}^{-1}\left\{\frac{A}{s+a}\right\}=Ae^{-at}$$

Distinct Real Poles: Cover-Up Method

Example 1. Derive the quick method first; use coefficient matching only as an alternative.

1 Choose the expansion

Each distinct linear factor gets one constant numerator. The original fraction is already strictly proper.

$$F(s) = \frac{s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

2 Multiply, cancel, then substitute

Multiply both sides by $(s+1)$. After cancellation, evaluate at $s=-1$ (the limit at the pole). The B term becomes zero, so $A=2$. Repeat with $(s+2)$ for B.

$$\times(s+1): \frac{s+3}{s+2} = A + \frac{B(s+1)}{s+2}$$

$$A = \left[\frac{s+3}{s+2}\right]_{s=-1} = \frac{-1+3}{-1+2} = 2$$

$$B = \left[\frac{s+3}{s+1}\right]_{s=-2} = \frac{-2+3}{-2+1} = -1$$

3 Alternative: match coefficients

Clear both denominators and expand. The two numerators agree for every s, so their s coefficients and constants must match. Solve the resulting two equations together.

$$s+3 = A(s+2) + B(s+1) = (A+B)s + (2A+B)$$

$$A+B=1, \quad 2A+B=3 \quad \Rightarrow \quad A=2, B=-1$$

4 Invert the result

A simple linear denominator gives an exponential. Keep both the coefficient and the sign of each term.

$$F(s) = \frac{2}{s+1} - \frac{1}{s+2} \quad \Rightarrow \quad f(t) = 2e^{-t} - e^{-2t}$$

Long Division Before Expansion

Example 2. A polynomial part must come out first when the numerator degree is not lower.

1 Compare the degrees

Both polynomials have degree two. A sum of simple fractions alone would miss the polynomial part.

$$F(s) = \frac{s^2 + 4s + 5}{(s+1)(s+2)} = \frac{s^2 + 4s + 5}{s^2 + 3s + 2}$$

2 Divide and subtract

The leading terms give quotient one. Subtract one copy of the denominator from the numerator to get the remainder.

$$(s^2 + 4s + 5) - (s^2 + 3s + 2) = s + 3$$

3 Expand the remainder

The remainder is now proper. At $s = -1$, cancelling $(s+1)$ gives $A = 2$; at $s = -2$, cancelling $(s+2)$ gives $B = -1$.

$$F(s) = 1 + \frac{s+3}{(s+1)(s+2)} = 1 + \frac{2}{s+1} - \frac{1}{s+2}$$

4 Match the polynomial term

The quotient 1 becomes an impulse $\delta(t)$, **not a constant signal**: $\mathcal{L}\{\delta(t)\} = 1$, whereas $\mathcal{L}\{1\} = \frac{1}{s}$.

$$\mathcal{L}\{\delta(t)\} = 1, \qquad \mathcal{L}\{1\} = \frac{1}{s}$$

5 Combine the time terms

Invert all three terms. Recombine the fractions to check that the original numerator returns.

$$f(t) = \delta(t) + 2e^{-t} - e^{-2t}$$

Quadratic Factors

Example 3. Use real coefficients instead of complex factors.

1 Complete the square

Read the shift and oscillation frequency from the denominator.

$$F(s) = \frac{2s+12}{s^2+2s+5} = \frac{2s+12}{(s+1)^2+2^2}$$

2 Match the numerators

Split the numerator to match the shifted cosine and sine pairs below.

$$2s+12 = 2(s+1)+10 = 2(s+1)+5\cdot 2$$

$$F(s) = 2\frac{s+1}{(s+1)^2+2^2} + 5\frac{2}{(s+1)^2+2^2}$$

3 Identify the two table pairs

Use the **same shift in numerator and denominator**.
The sine coefficient includes **division by the frequency**.

$$\frac{s+\alpha}{(s+\alpha)^2+\omega^2} \leftrightarrow e^{-\alpha t}\cos(\omega t)$$

$$\frac{\omega}{(s+\alpha)^2+\omega^2} \leftrightarrow e^{-\alpha t}\sin(\omega t)$$

4 Keep the outside coefficients

With $\alpha = 1$ and $\omega = 2$, the numerator 10 equals 5ω. The sine term therefore has coefficient 5.

$$f(t) = 2e^{-t}\cos(2t) + 5e^{-t}\sin(2t)$$

Repeated Poles

Example 4. Include every power of a triple pole.

1 Include all three powers

A triple factor needs three coefficients. **Include every lower power first**. A coefficient may later turn out to be zero.

$$F(s) = \frac{s^2 + 2s + 3}{(s+1)^3} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{(s+1)^3}$$

2 Clear the denominator

Multiply through by the full cubic factor. Each term keeps the powers left after cancellation.

$$s^2 + 2s + 3 = A(s+1)^2 + B(s+1) + C$$

3 Match powers and solve

Expand the right side: $As^2 + (2A + B)s + (A + B + C)$.
Match it with $s^2 + 2s + 3$: $A = 1$, then $B = 0$, then $C = 2$.

$$A = 1, \quad 2A + B = 2, \quad A + B + C = 3 \quad \Rightarrow \quad B = 0, C = 2$$

4 Apply the factorial rule

With $m = 3$, $\frac{1}{(s+1)^3}$ gives $\frac{t^2}{2}e^{-t}$. The coefficient 2 in $F(s)$ cancels the 2, leaving t^2e^{-t}.

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^m}\right\} = \frac{t^{m-1}}{(m-1)!}e^{-at}, \quad m = 1,2,\ldots$$

5 Combine and verify

The middle coefficient is **zero only after solving**. Recombine the remaining terms to check $F(s)$.

$$F(s) = \frac{1}{s+1} + \frac{2}{(s+1)^3} \quad \Rightarrow \quad f(t) = e^{-t} + t^2e^{-t}$$